

Particle System Model

Flux

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1 State and Dynamics

Let n particles evolve in $\mathbb{R}^2$ with positions

$$X(t) = (x_1(t), \dots, x_n(t)), \quad x_i(t) \in \mathbb{R}^2.$$

The motion with friction is

$$\ddot{x}_i + \alpha \dot{x}_i = -f_E(x_i) - f_I(X, x_i), \quad i = 1, \dots, n,$$

where $\alpha \geq 0$ is the friction coefficient.

2 External Force (Context-Free)

The external term depends only on the local position:

$$f_E(x) : \mathbb{R}^2 \rightarrow \mathbb{R}^2.$$

In this application, f_E is built from user-controlled radial centers and does not depend on the current full particle configuration X .

3 Internal Force (In-Context)

Internal interactions depend on the whole configuration X . We define:

Gaussian head kernel

$$g_s(r) = \exp\left(-\frac{\|r\|^2}{s^2}\right).$$

Unity On/Off vector term

$$w_{ij} = \begin{cases} \frac{x_i - x_j}{\|x_i - x_j\|}, & \text{Unity On,} \\ x_i - x_j, & \text{Unity Off.} \end{cases}$$

with the convention that the self-term $j = i$ contributes zero in implementation.

Interaction map For any query position $x \in \mathbb{R}^2$ and configuration $X = (x_1, \dots, x_n)$, define $w(x, x_j)$ by the same Unity rule above, and

$$f(x, X) : \mathbb{R}^2 \times (\mathbb{R}^2)^n \rightarrow \mathbb{R}^2.$$

Attention Off

$$f(x, X) = \frac{1}{n} \sum_{k=1}^K \sum_{j=1}^n v_k g_{s_k}(x - x_j) w(x, x_j).$$

Attention On

$$f(x, X) = \sum_{k=1}^K \sum_{j=1}^n v_k \frac{g_{s_k}(x - x_j)}{\sum_{\ell=1}^n g_{s_k}(x - x_\ell)} w(x, x_j).$$

The model equation uses

$$f_I(X, x_i) = f(x_i, X).$$

4 Two-Head Case ($K = 2$)

With two heads:

$$v_1 > 0 \quad (\text{repulsive}), \quad v_2 < 0 \quad (\text{attractive}).$$

Using UI amplitudes

$$a_R = |v_1|, \quad a_A = |v_2|,$$

we write

$$v_1 = +a_R, \quad v_2 = -a_A.$$

With radii s_R (repulsion) and s_A (attraction), the scalar radial profile is

$$\phi(r) = a_R \exp\left(-\frac{r^2}{s_R^2}\right) - a_A \exp\left(-\frac{r^2}{s_A^2}\right),$$

which is equivalent to the two-head expansion of $f(x, X)$ above.